

Assignment 1

00103335: Deep Learning and Reinforcement Learning

Fall 2026

Problem 1

(a) For $f = \sum_{j=1}^q a_j \phi_j$, write $f - m = \sum_{j=1}^Q c_j \phi_j$, where $c_j = a_j - \theta_j$ for $j \leq q$ and $c_j = -\theta_j$ for $j > q$. Orthonormality gives

$$\begin{aligned}\|f - m\|_{L^2(P_X)}^2 &= \sum_{j,k=1}^Q c_j c_k \mathbb{E}_X[\phi_j(X) \phi_k(X)] = \sum_{j=1}^Q c_j^2 \\ &= \sum_{j=1}^q (a_j - \theta_j)^2 + \sum_{j=q+1}^Q \theta_j^2.\end{aligned}$$

Thus the unique minimizer is $f_q^* = \sum_{j=1}^q \theta_j \phi_j$. All empty sums below are zero, covering $q = 0$.

(b) Since $\mathbb{E}\xi_j = 0$,

$$\mathbb{E}_S \widehat{f}_q(x) = \sum_{j=1}^q \left(\theta_j + \frac{\nu}{\sqrt{n}} \mathbb{E}\xi_j \right) \phi_j(x) = \sum_{j=1}^q \theta_j \phi_j(x) = f_q^*(x).$$

Consequently, by orthonormality,

$$B_q^2 = \sum_{j=q+1}^Q \theta_j^2, \quad B_{q+1}^2 - B_q^2 = -\theta_{q+1}^2 \leq 0.$$

(c) Independence and $\text{Var}(\xi_j) = 1$ imply $\text{Cov}(Z_j, Z_k) = \nu^2 \delta_{jk}/n$. Thus

$$\begin{aligned}\text{Var}_S(\widehat{f}_q(x)) &= \sum_{j,k=1}^q \phi_j(x) \phi_k(x) \text{Cov}(Z_j, Z_k) = \frac{\nu^2}{n} \sum_{j=1}^q \phi_j(x)^2, \\ V_q &= \frac{\nu^2}{n} \sum_{j=1}^q \mathbb{E}_X[\phi_j(X)^2] = \frac{q\nu^2}{n}.\end{aligned}$$

Hence $V_{q+1} - V_q = \nu^2/n > 0$.

Problem 2

(a) Expand the quadratic loss:

$$\widehat{L}(b) = \frac{1}{2n} (y^\top y - 2b^\top X^\top y + b^\top X^\top X b).$$

Since $X^\top X$ is symmetric, differentiation gives

$$\nabla \widehat{L}(b) = \frac{1}{n} X^\top (Xb - y), \quad b_{t+1} = b_t + \frac{\eta}{n} X^\top (y - Xb_t).$$

The Hessian is $\nabla^2 \widehat{L}(b) = X^\top X/n$. For any v , $v^\top X^\top X v = \|Xv\|_2^2$, so $\ker(X^\top X) = \ker X$ has dimension $d - n > 0$. Thus the Hessian is singular.

(b) Set $A = XX^\top$ and $M = I_n - \eta A/n$. Then

$$r_{t+1} = y - X \left(b_t + \frac{\eta}{n} X^\top r_t \right) = Mr_t, \quad r_0 = y, \quad r_t = M^t y.$$

Because X has full row rank, $A = U \operatorname{diag}(\lambda_i) U^\top$ with U orthogonal and $\lambda_i > 0$. Therefore

$$\|r_t\|_2^2 = \sum_{i=1}^n (1 - \eta \lambda_i/n)^{2t} (u_i^\top y)^2 \leq \rho^{2t} \|y\|_2^2, \quad \rho = \max_i |1 - \eta \lambda_i/n|.$$

For $0 < \eta < 2n/\lambda_{\max}(A) = 2n/\|X\|_{\text{op}}^2$, every absolute value is less than one. Hence $r_t \rightarrow 0$.

(c) The matrix geometric sum satisfies

$$(I_n - M) \sum_{s=0}^{t-1} M^s = \sum_{s=0}^{t-1} (M^s - M^{s+1}) = I_n - M^t, \quad (I_n - M)^{-1} = \frac{n}{\eta} A^{-1}.$$

Under the step-size condition in (b), summing the updates yields

$$\begin{aligned} b_t &= \frac{\eta}{n} X^\top \sum_{s=0}^{t-1} M^s y = \frac{\eta}{n} X^\top (I_n - M)^{-1} (I_n - M^t) y \\ &= X^\top (XX^\top)^{-1} \left[I_n - \left(I_n - \frac{\eta}{n} XX^\top \right)^t \right] y \longrightarrow b^\dagger := X^\top (XX^\top)^{-1} y. \end{aligned}$$

(d) $Xb^\dagger = AA^{-1}y = y$. Any interpolator b satisfies $X(b - b^\dagger) = 0$, so $b = b^\dagger + v$ with $v \in \ker X$. Also

$$(b^\dagger)^\top v = y^\top A^{-1} X v = 0, \quad \|b\|_2^2 = \|b^\dagger\|_2^2 + \|v\|_2^2.$$

The norm is minimized uniquely when $v = 0$.

(e) For arbitrary b_0 , $r_0 = y - Xb_0$. The same sum gives

$$b_t = b_0 + X^\top A^{-1} (I_n - M^t) (y - Xb_0) \longrightarrow b^\dagger + (I_d - X^\top A^{-1} X) b_0.$$

For $P = X^\top A^{-1} X$, direct calculation gives

$$P^\top = P, \quad P^2 = X^\top A^{-1} (XX^\top) A^{-1} X = P.$$

For $u \in \mathbb{R}^n$ and $v \in \ker X$,

$$P(X^\top u) = X^\top A^{-1} A u = X^\top u, \quad P v = X^\top A^{-1} X v = 0.$$

Hence P projects orthogonally onto $\operatorname{range}(X^\top)$. Thus $I_d - P = \Pi_{\ker X}$, proving the claimed limit.

Problem 3

Write $G = (X^\top X)^{-1}$, which is symmetric.

(a) $H^\top = (XGX^\top)^\top = H$ and $H^2 = XG(X^\top X)GX^\top = H$.

(b) If $z = Xa$, then $H z = XG(X^\top X)a = z$. If $X^\top z = 0$, then $H z = 0$. Conversely, $H z = 0$ implies $X^\top z = X^\top H z = 0$. Thus H is the orthogonal projector onto $\text{range}(X)$; $I_n - H$ projects onto its orthogonal complement $\ker(X^\top)$.

(c) These orthogonal subspaces have dimensions d and $n - d$, so the eigenvalues of H are 1 (multiplicity d) and 0 (multiplicity $n - d$). Consequently

$$\text{rank}(H) = \text{tr}(H) = d, \quad \text{tr}(I_n - H) = n - d.$$

For each coordinate vector e_i , $h_{ii} = e_i^\top H e_i = \|H e_i\|_2^2 \geq 0$ and $1 - h_{ii} = \|(I_n - H)e_i\|_2^2 \geq 0$. Also $\sum_i h_{ii} = \text{tr}(H) = d$.

(d) At a minimizer,

$$0 = \nabla \widehat{L}(\widehat{b}) = \frac{1}{n} X^\top (X\widehat{b} - y) \implies (X^\top X)\widehat{b} = X^\top y \implies \widehat{b} = GX^\top y.$$

The positive definite Hessian $X^\top X/n$ ensures this is the unique minimizer. Hence $\widehat{y} = Hy$ and $r = (I_n - H)y$. Since $X^\top (I_n - H) = 0$, $X^\top r = 0$. Moreover, $(Hy)^\top r = y^\top H(I_n - H)y = 0$, so

$$\|y\|_2^2 = \|Hy\|_2^2 + \|(I_n - H)y\|_2^2, \quad \min_b \frac{1}{2n} \|y - Xb\|_2^2 = \frac{1}{2n} \|r\|_2^2 = \frac{1}{2n} y^\top (I_n - H)y.$$

(e) Since $HX = X$, $(I_n - H)X\beta = 0$. Put $K = I_n - H$; $K^\top K = K$ and $Ky = K\eta$. Thus

$$\begin{aligned} \mathbb{E}[\|(I_n - H)y\|_2^2 \mid X] &= \mathbb{E}[\eta^\top K \eta \mid X] \\ &= \sum_{i,j=1}^n K_{ij} \mathbb{E}[\eta_i \eta_j \mid X] = \tau^2 \sum_{i,j=1}^n K_{ij} \delta_{ij} \\ &= \tau^2 \text{tr}(K) = \tau^2(n - d). \end{aligned}$$

Problem 4

(a) Conditional on the preceding columns, each new Gaussian column has a density and therefore has probability zero of belonging to their proper linear span in $\mathbb{R}^m$. Hence Z has rank p almost surely. For $a \neq 0$, $a^\top W a = \|Za\|_2^2 > 0$ almost surely.

(b) For $p > 1$, let Q permute column j to the first position, so $ZQ = [z_j, Z_{-j}]$. Then

$$\widetilde{W} = Q^\top W Q = \begin{pmatrix} z_j^\top z_j & z_j^\top Z_{-j} \\ Z_{-j}^\top z_j & Z_{-j}^\top Z_{-j} \end{pmatrix}.$$

Since $\widetilde{W}^{-1} = Q^\top W^{-1} Q$, $[\widetilde{W}^{-1}]_{11} = [W^{-1}]_{jj}$. The Schur complement of $Z_{-j}^\top Z_{-j}$ is

$$S = z_j^\top z_j - z_j^\top Z_{-j} (Z_{-j}^\top Z_{-j})^{-1} Z_{-j}^\top z_j = z_j^\top (I_m - P_{-j}) z_j.$$

The Schur-complement inverse formula gives

$$[W^{-1}]_{jj} = [\widetilde{W}^{-1}]_{11} = S^{-1} = (z_j^\top (I_m - P_{-j}) z_j)^{-1}.$$

For $p = 1$, this is the scalar inverse with $P_{-j} = 0$.

(c) Conditional on Z_{-j} , $I_m - P_{-j}$ is an orthogonal projector of rank $k = m - (p - 1) = m - p + 1$. Thus, for an orthogonal U ,

$$I_m - P_{-j} = U \operatorname{diag}(I_k, 0) U^\top.$$

Conditional on Z_{-j} , the independent $z_j \sim N(0, I_m)$ satisfies $g = U^\top z_j \sim N(0, U^\top U) = N(0, I_m)$. Therefore

$$z_j^\top (I_m - P_{-j}) z_j \mid Z_{-j} \stackrel{d}{=} \sum_{\ell=1}^k g_\ell^2 \sim \chi_k^2, \quad g_\ell \stackrel{\text{iid}}{\sim} N(0, 1).$$

(d) For $k > 2$, substitute $u = 2v$, $du = 2 dv$:

$$\begin{aligned} \mathbb{E}[U^{-1}] &= \frac{1}{2^{k/2} \Gamma(k/2)} \int_0^\infty u^{k/2-2} e^{-u/2} du \\ &= \frac{2^{k/2-1}}{2^{k/2} \Gamma(k/2)} \int_0^\infty v^{k/2-2} e^{-v} dv = \frac{\Gamma(k/2 - 1)}{2 \Gamma(k/2)} \\ &= \frac{1}{2(k/2 - 1)} = \frac{1}{k - 2}, \end{aligned}$$

where $\Gamma(k/2) = (k/2 - 1) \Gamma(k/2 - 1)$. For $0 < k \leq 2$, the integrand behaves as $u^{k/2-2}$ near zero, so the integral diverges. Here $k = m - p + 1 > 2$; by (b)–(c),

$$\mathbb{E}[[W^{-1}]_{jj}] = \mathbb{E}\left[\mathbb{E}\left[(z_j^\top (I_m - P_{-j}) z_j)^{-1} \mid Z_{-j}\right]\right] = \frac{1}{m - p - 1}.$$

(e) Put $B = W^{-1}$. Positive definiteness gives $|B_{rs}| \leq \sqrt{B_{rr} B_{ss}}$, so Cauchy–Schwarz and (d) imply

$$\mathbb{E}|B_{rs}| \leq \sqrt{\mathbb{E} B_{rr} \mathbb{E} B_{ss}} = \frac{1}{m - p - 1} < \infty.$$

For $r \neq s$, let D be diagonal with $D_{rr} = -1$ and all other diagonal entries 1. The Gaussian sign symmetry gives $ZD \stackrel{d}{=} Z$, hence $DBD \stackrel{d}{=} B$. Its (r, s) entry is $-B_{rs}$, so $\mathbb{E} B_{rs} = 0$. Together with (d), this proves

$$\boxed{\mathbb{E}[W^{-1}] = \frac{I_p}{m - p - 1}.$$